

Autonomous Vehicle Safety and Legal Compliance in Private Distribution Centers: A North American Analysis

1 Introduction

This report provides a comprehensive analysis of the legal and safety requirements for autonomous vehicles (AVs) and regular vehicles operating exclusively within private distribution center parking lots in Canada and the USA. The objective is to delineate specific legal mandates, identify pertinent safety standards, and propose best practices for ensuring safety during proof-of-concept testing in such controlled industrial environments. A key focus is to differentiate between public road regulations and the frameworks governing private property and workplace safety.

Autonomous vehicles are increasingly being integrated into controlled environments like distribution centers to enhance operational efficiency, reduce human error, and improve overall safety in repetitive tasks. Unlike public roads, these private operational domains present distinct regulatory challenges and opportunities. The legal and safety oversight for vehicles in these settings often falls under occupational health and safety (OHS) regulations rather than traditional traffic laws, necessitating a different approach to compliance and risk management.

2 Legal Requirements for Autonomous Vehicles in Private Parking Lots

This section examines whether an autonomous vehicle confined to a distribution center parking lot legally needs to adhere to specific safety standards. It provides references to relevant federal or state/provincial laws and highlights the differences between Canadian and U.S. policies.

2.1 Canada

In Canada, the legal landscape for vehicles operating on private property is primarily shaped by occupational health and safety legislation. Provincial traffic laws generally do not extend to vehicles operating exclusively on private property. For example, Ontario's Highway Traffic Act explicitly limits its regulations to "public roads and highways." Consequently, a vehicle operating solely within a private distribution center parking lot would typically not be subject to standard traffic infractions, unless a criminal offense, such as driving while impaired, is committed. This creates a regulatory gap for AV-specific operational rules under traffic law, effectively shifting the primary burden

of safety oversight onto other legal frameworks.

At the federal level, Transport Canada establishes safety regulations for vehicles sold within or imported into Canada under the Motor Vehicle Safety Act (Canada). This act sets standards for vehicle design and equipment, such as mandatory daytime running lights. This indicates that even if an AV never leaves private property, its fundamental design and safety features might still be subject to federal manufacturing standards if it is considered a "motor vehicle" for sale or import.

Occupational Health and Safety (OHS) legislation is the primary legal framework governing vehicle safety in private industrial settings. Both federal (Canada Labour Code Part II) and provincial OHS acts impose significant legal duties on employers. The Canada Labour Code Part II mandates employers to prevent workplace accidents and grants employees rights such as the "right to refuse dangerous work." Provincially regulated workplaces, which include most distribution centers, have similar legal duties. For instance, in Ontario, employers must protect workers from occupational driving hazards by ensuring vehicles are maintained to accepted standards and that safe driving policies are developed. This implies that the technical maintenance standards from public road legislation are incorporated by reference into workplace safety requirements.

The pervasive and legally binding nature of OHS and the Canada Labour Code confirms that the absence of specific AV traffic laws on private property is effectively covered by a comprehensive workplace safety framework. This means that the design, operation, and maintenance of the autonomous vehicle must meet general machinery and mobile equipment safety standards, necessitating robust risk assessments and safety management systems.

2.2 United States

In the United States, the legal requirements for autonomous vehicles on private property largely mirror Canada's approach. State traffic laws are generally not enforceable on private property, so an AV operating solely within a private distribution center parking lot would not be subject to standard state traffic violations. Exceptions exist for dangerous behaviors like reckless driving or DUI. The general non-applicability of state traffic laws on private property means that U.S. state-level AV legislation would not directly regulate an AV confined to a private distribution center.

Federal regulations, primarily through the National Highway Traffic Safety Administration (NHTSA), largely focus on motor vehicle safety for public roads. While federal AV guidance documents are generally voluntary, NHTSA's incident reporting

requirements for vehicles with Automated Driving Systems (ADS) or Level 2 (or higher) Advanced Driver Assistance Systems (ADAS) suggest that incidents on private property could fall under reporting expectations. This implies that maintaining robust incident reporting mechanisms is a crucial best practice.

The Occupational Safety and Health Administration (OSHA) provides the most direct and legally binding framework for vehicle safety in private industrial settings in the USA. OSHA's Standard 1926.601, "Motor vehicles," explicitly applies to "vehicles that operate within an off-highway jobsite, not open to public traffic," directly covering the distribution center parking lot scenario. This standard serves as the cornerstone of legal safety requirements, and its detailed provisions mean an AV must comply with its requirements. Furthermore, OSHA requires that electrical devices and cables in workplaces meet relevant UL standards.

3 Applicable Safety Standard Documents for Private Parking Lot Operation

Based on the analysis of legal applicability, this section identifies the safety standards documents that an autonomous vehicle would legally need to follow, as well as those that are highly recommended by industry consensus for robust safety.

3.1 Legally Mandated Standards

The legal requirements for autonomous vehicles on private property are primarily derived from workplace safety legislation. The AV must be treated as industrial machinery or mobile equipment, subject to rigorous safety standards for its design, maintenance, and interaction with the workplace environment.

United States

- **OSHA 1926.601 - Motor Vehicles:** This standard is explicitly applicable to vehicles operating in private industrial settings. Key requirements for an AV include:
 - Maintaining operable brake systems.
 - Equipping the vehicle with adequate lights and an audible warning device.
 - Implementing a reverse signal alarm or using an observer when backing up with an obstructed view.
 - Equipping vehicles **with cabs** with operable windshields and wipers.
 - Having seat belts and anchorages that meet Federal Motor Vehicle Safety Standards (FMVSS).
 - Performing pre-shift checks to ensure safe operating conditions.
- **UL Standards (via OSHA):** OSHA legally requires that almost all electrical

devices and cables in workplaces meet the relevant UL standards. Therefore, an AV's electrical components must comply with applicable UL standards to meet these legal requirements.

Canada

- **Federal and Provincial OHS Acts:** These acts legally mandate a safe working environment and the safe operation of mobile equipment. While less prescriptive than OSHA, they require that vehicles be maintained to accepted standards, sometimes referencing public road maintenance regulations.
- **Ontario Regulation 851 - Industrial Establishments:** This regulation legally mandates the use of "barriers, warning signs or other safeguards" to protect workers where vehicle or pedestrian traffic may endanger them.
- **Federal Motor Vehicle Safety Act (Canada):** If the AV is classified as a "motor vehicle" under federal law, it must comply with federal safety standards for its basic design.

3.2 Highly Recommended Industry Consensus Standards

Adherence to industry consensus standards is critical for demonstrating due diligence, managing liability, and ensuring a high level of safety, even though they are generally voluntary.

- **ISO 21448 (SOTIF):** This standard addresses "Safety of the Intended Functionality," focusing on hazards from system limitations rather than hardware or software failures. It emphasizes scenario-based testing and defining Operational Design Domains (ODD).
- **ISO 12100:** This foundational standard provides a systematic framework for identifying hazards, assessing risks, and implementing risk reduction measures for machinery.
- **ISO 26262:** A standard for functional safety of electrical and electronic (E/E) systems, it is critical for preventing hazards caused by system malfunctions. The standard's principles provide a robust framework for managing safety throughout the lifecycle of safety-critical components.
- **UL 4600:** This is a comprehensive standard for "fully autonomous products," including autonomous vehicles in "automated warehouse settings." It covers safety case construction, risk analysis, testing procedures, and autonomy validation.
- **SAE J3016:** This standard provides a globally recognized taxonomy for defining the capabilities of an autonomous vehicle, from Level 0 to Level 5.

Understanding the AV's SAE level is crucial for defining its ODD and the scope of necessary safety measures.

- **CSA Z432:** This Canadian standard applies to the "protection of workers from the hazards arising from the use of mobile or stationary machinery." It provides criteria for the selection and application of guards and protective devices.

4 Essential Safety Standard Sections for Proof-of-Concept Testing

Ensuring the safety of a proof-of-concept autonomous vehicle during testing in a private parking lot is paramount. Adhering to key sections of relevant safety standards establishes a robust safety framework and mitigates risks.

4.1 General Principles of Safety and Risk Management

A comprehensive risk assessment is a fundamental step. Standards like ISO 12100 and UL 4600 provide principles for hazard identification and risk reduction. It is recommended to conduct a Hazard Analysis and Risk Assessment (HARA), tailored to the specific distribution center, and systematically apply the hierarchy of controls: elimination, substitution, engineering controls, awareness means, administrative controls, and Personal Protective Equipment (PPE).

The concept of Safety of the Intended Functionality (SOTIF), detailed in ISO 21448, is crucial as it addresses risks from "functional insufficiencies" and "system limitations." This involves clearly defining the Operational Design Domain (ODD) and utilizing scenario-based testing to evaluate the vehicle's performance in both expected and unforeseen conditions.

4.2 Specific Technical and Operational Safety Requirements

- **Emergency Stop Systems:** AVs must be equipped with easily accessible and reliable emergency stop mechanisms. This includes both onboard and remote shutdown capabilities.
- **Collision Avoidance and Perception Systems:** The AV's perception system must accurately detect and classify objects under varying conditions. Standards like ISO 17757 and UL 4740 address perception systems and the functional safety of Lidar.
- **Operational Design Domain (ODD) Definition:** A clear definition of the ODD is essential, outlining the specific conditions under which the AV is designed to operate safely. For a proof-of-concept, the ODD should be precisely delimited to the private parking lot.
- **Verification and Validation (V&V):** Rigorous V&V is indispensable and includes

computer simulations, private track testing, and scenario-based testing to ensure the system operates safely in all conditions.

- **Human-Machine Interface (HMI):** Even in autonomous systems, a clear HMI is crucial for supervision, intervention, and communication. This includes mechanisms for a human operator to take over control.
- **Maintenance, Inspection, and Documentation:** Consistent maintenance, regular pre-operational inspections, and comprehensive documentation are critical for ongoing safety. A system for defect reporting and securing unattended equipment must be in place.

5 Conclusions

The legal and safety landscape for autonomous vehicles operating in private distribution center parking lots in Canada and the USA is distinct from public roads. The primary legal framework shifts to Occupational Health and Safety (OHS) legislation in both countries. In the U.S., OSHA's Standard 1926.601 provides a specific and legally binding framework. In Canada, federal and provincial OHS acts impose a general duty of care on employers.

For a proof-of-concept AV, adherence to industry consensus standards like ISO 21448 (SOTIF), ISO 26262 (Functional Safety), and UL 4600 (Evaluation of Autonomous Products) is critical for demonstrating due diligence and ensuring safety. These standards provide the detailed technical and procedural guidance that general OHS laws imply but do not explicitly state. Proactive adherence to these standards is essential for mitigating risks and fostering confidence in the deployment of this technology.